

CUSTOMER MILESTONE BRIEF

MS #33 - Database Semantic Core

Purpose, customer value, delivered capability, evidence, and claim boundaries

Status	Complete
Release	0.33.0
Date	2026-08-30
Audience	Customers, partners, executives, architects, delivery teams, and assurance reviewers

Executive summary

MS #33 - Database Semantic Core converts a specific part of the LIGHTYEAR modernization approach into a governed, reviewable capability. Creates the platform-neutral semantic contract needed to compare database behavior instead of merely generating target DDL.

Why this matters to a customer

Creates the platform-neutral semantic contract needed to compare database behavior instead of merely generating target DDL.

The practical result is a narrower, evidence-backed decision instead of a broad modernization claim. Commercial, architecture, delivery, security, and assurance teams can review the same capability, proof, and limits.

What the milestone delivers

- Added independent source- and target-adapter interfaces plus a 17-kind canonical database type system with explicit numeric, null, time-zone, and truncation rules.
- Added deterministic data-profile and schema-transformation contracts, typed normalized rows, query/result comparison, transaction comparison, a content-bound CDC envelope, and fail-closed cutover and rollback contracts.
- Added a content-addressed AUTHFRDS compatibility ledger covering every column for PostgreSQL and Oracle plus transaction isolation, DDL CDC, sequence state, and stored logic.
- Required exactly one of exact, normalized-equivalent, policy-decision-required, lossy, or unsupported for every ledger entry; unresolved policy or loss blocks equivalence and unsupported behavior is excluded from the claim scope.

- Added a deterministic adapter conformance suite and committed receipt for the existing PostgreSQL and Oracle adapters, with explicit non-promotion and non-production boundaries.
- Added frozen JSON schemas and tamper tests for semantic contracts, normalized rows, CDC events, compatibility coverage, policy decisions, query/transaction behavior, and cutover/rollback gates.

How it differs from earlier work

MS #33 builds on MS #32 (Governed Pilot Work Package). The earlier milestone established its own bounded capability; this milestone adds database semantic core while preserving the earlier evidence and its limits.

Earlier evidence is not automatically promoted into this milestone. A parser, simulator, local comparison, or generated artifact is not live platform equivalence or production readiness.

What a customer can do with it

A customer can use database semantic core as a bounded decision and delivery capability, attached to approved sources, owners, policies, target architecture, acceptance criteria, and authorized evidence.

Unresolved semantic loss, policy decisions, unsupported behavior, and live-evidence requirements remain visible.

Evidence and acceptance posture

The repository capability is complete because its committed implementation and release record are present. Customer qualification remains claim-specific and evidence-class-specific.

Review exact source and artifact identities, distinguish development from authorized live evidence, inspect policy and loss classifications, confirm passed and blocked gates, and verify that later changes have not invalidated the evidence.

Boundaries and claims not made

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- Added a deterministic adapter conformance suite and committed receipt for the existing PostgreSQL and Oracle adapters, with explicit non-promotion and non-production boundaries.

Source of record

- CHANGELOG.md release section(s): 0.33.0
- README.md product and operating guidance
- LIGHTYEAR-ROADMAP.md governing sequence and claim classifications

- docs/milestones/catalog.json metadata and docs/milestones/manifest.json artifact integrity

The underlying schemas, receipts, ledgers, tests, and policy decisions remain authoritative when greater detail is required.